

YASHASWINI GOWDA M

Phone: 91+ 8310777365 Mail: yashaswini11092005@gmail.com [linkedin.com/in/yashaswini-gowda-m/](https://www.linkedin.com/in/yashaswini-gowda-m/)
github.com/yashaswini-1128

Education

Year	Degree	Institute	CGPA / Percentage
2023 – 2027	B. Tech – Computer Science & Engineering	Dayananda Sagar University (DSU)	8.66 / 10 (Till 5 th Sem)
2021 – 2023	12 th Grade Karnataka State Board – PCMB	Christ Academy Junior School (CAJC)	85.5 %
2011 – 2021	10th Grade – CBSE	New Baldwin International School (NBIS)	70.5%

Projects

Explainable CNN Framework for Automated MRI Distortion Detection | *Python, TensorFlow, Keras, Grad-CAM, OpenCV*

- Designed a CNN model to classify MRI scans as good or distorted, achieving **94.5% accuracy** and **95.2% recall**.
- Simulated real-world artifacts using multi-intensity Gaussian noise and blur to train a robust distortion detection pipeline.
- Applied Grad-CAM to generate heatmaps highlighting distortion-relevant regions, improving model transparency for clinical use.
- Co-authored and published a research paper on Explainable AI for medical imaging at Dayananda Sagar University.

Niyantrana SOS – Tactical Operations Dashboard | *Next.js, TypeScript, Firebase, React*

- Built a **real-time tactical HUD** displaying live unit movements and incidents using **Next.js App Router**.
- Engineered custom hooks for API polling, unit animation, and incident state management.
- Integrated Firebase for real-time data sync across multi-route tactical and dashboard views.
- Configured environment-based Firebase setup and deployed production build on **Vercel**.
- Structured scalable codebase with reusable Map, HUD components and TypeScript throughout.

RockShield AI – Rockfall Prediction & Alert System | *Python, TensorFlow, PyTorch, Scikit-learn, React.js, Django*

- Built an AI-driven web tool to predict rockfall risks using geological inputs (slope, soil, depth) and live weather data via **OpenWeatherMap API**.
- Developed ML models using Scikit-learn, PyTorch, and TensorFlow to classify risk levels on a color-coded dashboard.
- Implemented a Digital Twin simulation of mine walls, continuously updated with real-world sensor and drone data.
- Integrated automated alerts via Twilio/SMTP for evacuation instructions and blasting delay recommendations.
- Shortlisted for Smart India Hackathon (SIH) 2025 under Disaster Management theme (**Problem ID: SIH25071**).

IBM Cloud & AI – Emerging Technologies | *IBM Cloud, Watson AI, ML Deployment*

- Gained hands-on experience in Artificial Intelligence and Cloud Computing using the IBM Cloud Platform.
- Explored IBM Cloud services including Watson AI, Cloud Object Storage, and Machine Learning model deployment pipelines.
- Completed guided projects and assignments showcasing skills in data visualization, model integration, and cloud infrastructure setup.
- Certified through the Skills Build program by IBM and the **Edunet Foundation**.

Technical Skills & Certifications

Languages: Python, Java, C, HTML, CSS, JavaScript

Technologies / Frameworks: Machine Learning, Data Preprocessing, Full Stack Development, GitHub, Power BI, Data Visualization, UI/UX Design

Certifications: Python Course Completion- Geeks-of-geeks, Skills build and IBM cloud- Edunet foundation, Fundamental Of PowerBI